

BALEA LAC, Romania

WMO: 152790

Lat: 45.6039N

Lon: 24.6147E

Elev: 2071

StdP: 78.79

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-16.8	-28.9	0.3	-8.4	-25.3	0.5	-9.3	15.2	-4.4	13.2	-4.1	2.6	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	17.0	11.9	15.7	11.5	14.6	11.0	13.5	15.6	12.6	14.6	11.8	13.8	2.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	11.8	14.4	11.8	11.1	13.6	11.0	10.5	12.9	45.1	15.7	42.3	14.6	40.1	13.9	17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	10.9	9.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-7.2	-5.7	-4.2	0.1	4.2	8.3	9.4	10.6	6.2	2.9	-2.0	-5.5		
	DBStd	7.50	4.99	4.58	4.73	4.63	3.23	3.56	3.37	3.28	4.32	4.77	5.09	4.93		
	HDD10.0	3241	534	438	441	298	183	72	52	34	124	224	360	482		
	HDD18.3	6159	793	672	699	548	439	299	276	240	363	480	610	740		
	CDD10.0	123	0	0	0	0	1	22	34	52	11	2	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	2.9	3.1	3.8	3.1	3.4	2.5	2.6	2.7	2.3	2.8	2.6	3.3	3.2		
(15) Precipitation	PrecAvg	982	56	51	61	77	107	117	133	100	84	72	62	57		
	PrecMax	1605	118	124	110	195	259	224	234	208	178	148	164	117		
	PrecMin	602	14	7	19	20	46	14	18	18	11	1	2	7		
	PrecStd	263	30	28	25	40	55	54	60	58	42	40	40	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.9	8.0	11.0	13.4	17.9	18.9	18.8	18.1	13.9	9.2	9.3		
		MCWB	-1.1	1.2	2.8	5.3	9.1	10.7	11.1	13.1	10.6	5.4	4.3	0.5		
	2%	DB	1.7	3.4	5.5	9.1	11.6	15.9	16.7	16.8	14.5	11.9	7.1	4.7		
		MCWB	-2.2	0.4	2.2	4.7	7.6	11.4	12.0	12.3	10.3	4.5	1.9	-0.6		
	5%	DB	0.2	1.3	3.9	7.6	10.2	14.6	15.2	15.7	13.0	10.0	5.5	2.5		
		MCWB	-2.1	-1.1	0.9	3.7	6.8	10.7	11.7	11.7	9.4	4.3	1.6	-0.6		
	10%	DB	-0.9	-0.1	1.9	6.3	8.7	13.0	13.9	14.6	11.4	8.7	4.1	0.6		
		MCWB	-2.6	-2.3	0.0	3.0	6.5	10.2	11.0	11.2	8.0	4.4	1.6	-1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	2.7	4.6	6.9	10.2	14.1	14.4	14.5	12.4	8.6	5.7	3.2		
		MCDB	2.0	4.4	6.8	9.4	12.0	15.5	16.4	16.8	15.5	10.6	8.2	4.4		
	2%	WB	-0.6	0.8	2.7	5.6	8.5	12.3	13.2	13.7	11.3	7.1	4.2	1.6		
		MCDB	0.5	3.5	4.6	8.1	10.2	14.3	15.2	15.9	14.2	9.5	5.7	3.9		
	5%	WB	-1.2	-0.6	1.5	4.5	7.6	11.5	12.4	12.8	9.8	6.2	3.4	0.0		
		MCDB	-0.1	1.1	3.4	7.3	9.3	13.5	14.1	14.7	12.3	8.8	4.5	1.7		
	10%	WB	-2.2	-1.4	0.1	3.5	6.7	10.7	11.6	12.0	8.8	5.2	2.4	-1.1		
		MCDB	-1.2	-0.2	1.8	6.1	8.4	12.8	13.3	14.1	11.0	7.7	3.4	0.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.6	6.0	5.2	4.6	4.7	4.2	4.0	3.9	5.0	5.1	5.2		
		MCDBR	5.7	6.7	7.7	6.3	6.4	6.5	5.9	4.9	4.6	6.4	5.6	6.5		
	5% WB	MCWBR	4.0	4.3	5.1	4.2	5.6	5.2	5.3	4.5	4.6	5.1	3.8	3.8		
		MCWBR	4.9	5.7	6.4	5.8	6.0	5.6	4.9	4.3	4.3	5.7	3.9	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.234	0.253	0.280	0.318	0.310	0.312	0.322	0.328	0.300	0.276	0.259	0.236		
	taud		2.455	2.436	2.461	2.422	2.471	2.481	2.444	2.421	2.498	2.572	2.571	2.500		
	Ebn at Noon		908	948	960	943	957	953	938	921	925	910	872	870		
	Edh at Noon		66	83	94	108	106	106	109	107	90	72	60	57		
(44) All-Sky Solar Radiation	RadAvg		1.39	2.21	3.28	4.27	5.21	5.63	5.72	5.19	3.79	2.63	1.54	1.13		
	RadStd		0.14	0.28	0.35	0.47	0.51	0.51	0.50	0.44	0.38	0.34	0.16	0.13		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(47) Regional (77 neighbors)	+0.62	N/A	N/A	+0.78	N/A	N/A	-124	-175	+80	+25		

Nomenclature: See separate page